Programming Fundamentals- Fall 2025

Project Title

Thread Dyeing Management System

(TDMS)

Version 1.0

Submitted By: 2025-CS-79 (Section B)

Instructor: Mr. Kamran Mustafa

Department of Computer Science

University of Engineering and Technology Lahore

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1 Project Details

1.1 Project Summary

The **Textile Dyeing Management System (TDMS)** is a comprehensive management tool designed to streamline the operations of a textile dyeing factory. In a traditional setting, production records, worker attendance, and financial invoices are often maintained manually, leading to data loss and calculation errors. This application digitizes these processes, providing a centralized platform for both administrators (factory owners) and users (workers/employees).

The project is built on the core principle of high-dimensional data tracking. It uses a **3D Array structure** to store work production data across three axes: Employees, Parties (Clients), and Days of the month. This allows the system to generate highly specific reports, such as exactly how many bundles a specific worker dyed for a specific client on a specific day.

For the Administrator, the system offers powerful tools to manage the workforce (adding/removing employees, setting wage rates), manage clients (updating party lists and bundle prices), and tracking overhead factory expenses. The most significant feature is the **Automated Financial Report**, which generates a professional HTML document. This report calculates "Net Business Profit" by weighing total revenue against expenses and employee salaries, including automatic deductions for worker absenteeism.

For the Workers, the system provides a simplified portal to log their daily output and view their weekly progress. This transparency ensures that workers are paid accurately for the work they perform. The system ensures data persistence by saving all information into flat-text databases (`.txt` files), allowing the factory to maintain records over long periods.

1.2 Project Features

- **Dual-Layer Authentication:** Secure login system for Admin and Workers with role-based access control.
- **3D Production Tracking:** Records daily bundle counts for multiple parties across a 31-day cycle.
- **Dynamic Payroll Engine:** Calculates salaries based on individual piece-rates and applies penalties for unexcused absences.
- **Party & Client Management:** Real-time updates for client lists and market prices per bundle.
- **Expense Tracking:** Categorized logging of factory overheads (electricity, chemicals, maintenance).
- **Automated HTML Reporting:** Generates a visually styled financial statement using CSS, accessible via any web browser.
- **Data Persistence:** Automatic saving and loading of all system states using file handling.

1.3 Main Menu

1.3.1 Sign Up (Create New Account)

1.3.1.1 Purpose of The Menu

To allow new administrators or staff members to register themselves in the system. This adds their credentials to the permanent database so they can log in later.

1.3.1.2 Input/Output

- **Input:** User enters a new `Username` (string), a `Password` (string).
- **Output:** The system adds the data to the `users.txt` file and displays a message: "Registration Successful! You can now login."

1.3.1.3 Validation

The system checks if the `userCount` has reached the `size` limit (100). It also ensures that the `Username` is not already taken by another user in the `nameArray`.

1.3.2 Login

1.3.2.1 Purpose of The Menu

To authenticate users and direct them to their respective dashboards based on their assigned roles (Admin or User).

1.3.2.2 Input/Output

- **Input:** String `username`, String `password`.
- **Output:** Redirects to `adminInterface()` if the role is Admin, or `userInterface()` if the role is User. Displays "Invalid Credentials" on failure.

1.3.2.3 Validation

The system iterates through the `nameArray` and `passwordArray`. If no match is found, it prompts the user to try again. Empty inputs are not accepted.

1.3.3 Exit System

1.3.3.1 Purpose of The Menu

To safely terminate the application. Before closing, the system ensures all current session data (attendance, work records, and user lists) is securely written to the text files.

1.3.3.2 Input/Output

- **Input:** User selects the "Exit" option from the main menu.
- **Output:** The `saveData()` function is triggered to update all `.txt` files, and a "Thank you for using TDMS" message is displayed before the console closes.

1.3.3.3 Validation

The system asks for a final confirmation (Y/N) or simply executes the `saveData()` call to prevent any data loss from the current session.

1.3.3.4 Screenshot

-Main Menu:

```
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1.4 Admin Sub-Menu

1.4.1 2.2.1 Party Module (Client Management)

1.4.1.1 Purpose of The Menu

To manage the clients the factory works with. This includes adding new "Parties," viewing existing ones, and updating the price paid per bundle for dyeing.

1.4.1.2 Input/Output

- **Input:** Admin enters choice (1-4). To add, inputs `Party Name` (String) and `Bundle Price` (Int).
- **Output:** A list of all parties with their respective prices displayed in a table.

1.4.1.3 Validation

Ensures the system doesn't exceed `maxParties`. If a user enters a character instead of a number for the price, the system uses a validation loop to ask for a valid integer.

-Admin Sub-Menu:

-Party module:

```
|-----------------------------------------------------------------------------------------------
|       Main menu > Login > Admin Panel > Party Management
|-----------------------------------------------------------------------------------------------


|*|......~~~~~~~~___~_~_~_/\/\/\( Party Management Module )/\/\/\_~_~_~_~___~~~~~~~~~.......|*|
|*|                                                                                          |*|
|*|    1. ADD A PARTY                                                                        |*|
|*|    2. SEARCH A PARTY                                                                     |*|
|*|    3. UPDATE A PARTY                                                                     |*|
|*|    4. DELETE A PARTY                                                                     |*|
|*|    0. EXIT                                                                               |*|
|*|                                                                                          |*|
|*|       ENTER YOUR CHOICE...                                                               |*|
|*|__________________________________________________________________________________________|*|
Enter your Choice...
```

1.4.2 2.2.2 Employee Module (Worker Management)

1.4.2.1 Purpose of The Menu

To handle the factory workforce. The Admin can register new employees, set their "Piece-Rate" wages, or remove employees who are no longer working.

1.4.2.2 Input/Output

- **Input:** Admin enters `Employee Name` (String) and `Wage Rate` (Int).
- **Output:** Updates the `employee.txt` file and refreshes the internal `employeeName` array.

1.4.2.3 Validation

Checks if the employee name already exists to prevent duplicates. It also validates that the wage rate is a positive number.

-Employee Module:

```
|-----------------------------------------------------------------------------------------------
|       Main menu > Login > Admin Panel > Employee Management
|-----------------------------------------------------------------------------------------------


|*|......~~~~~~~~__~_~_~_/\/\/\( Employee Management Module )/\/\/\_~_~_~_~__~~~~~~~~~.......|*|
|*|                                                                                          |*|
|*|    1. ADD AN EMPLOYEE                                                                    |*|
|*|    2. SEARCH AN EMPLOYEE                                                                 |*|
|*|    3. UPDATE AN EMPLOYEE                                                                 |*|
|*|    4. DELETE AN EMPLOYEE                                                                 |*|
|*|    0. EXIT                                                                               |*|
|*|                                                                                          |*|
|*|       ENTER YOUR CHOICE...                                                               |*|
|*|__________________________________________________________________________________________|*|
Enter your Choice...
```

1.4.3 2.2.3 Mark Attendance

1.4.3.1 Purpose of The Menu

To record the daily presence of workers. This data is critical for the payroll system to calculate "Absent Penalties" during the financial report generation.

1.4.3.2 Input/Output

- **Input:** Admin enters the `Day` (1-31). For each employee, the Admin enters `1` for Present or `0` for Absent.
- **Output:** Saves a 2D matrix into `attendance.txt`.

1.4.3.3 Validation

Ensures the `Day` is between 1 and 31. If the Admin enters any number other than 0 or 1 for attendance, the system prompts for a correction.
-Marking attendance:

```
|-------------------------------------------------------------------------------------------
|       Main menu > Login > Admin Panel > Mark Attendance
|-------------------------------------------------------------------------------------------
Enter Day to mark attendance (1-31): 1
Is Ali Present? (y/n): y
Is Awais Present? (y/n): y
Is Hamza Present? (y/n): y
Attendance marked successfully!Press Any Key to Continue..
```

1.4.4 2.2.4 Factory Expense Tracking

1.4.4.1 Purpose of The Menu

To log overhead costs such as electricity bills, chemical purchases, and machinery maintenance. These expenses are subtracted from revenue to calculate net profit.

1.4.4.2 Input/Output

- **Input:** `Expense Date` (String), `Category` (String), and `Amount` (Int).
- **Output:** The expense is added to the `expenseAmount` array and saved to `expenses.txt`.

1.4.4.3 Validation

The system validates that the `Amount` is a numeric value. It ensures that the total number of expenses doesn't exceed the array limit.

-Factory Expense Tracker:

```
|-----------------------------------------------------------------------------------------
|       Admin Panel > Factory Expense Tracker
|-----------------------------------------------------------------------------------------

1. Add New Expense
2. View Detailed Log
3. View Category Summary
0. Back
Enter your Choice...1
Description (e.g. Rent): Rent
Category (Utility/Material/Salary): Utility
Date (DD/MM): 25/12
Amount: 50
Expense logged successfully!
```

```
|-----------------------------------------------------------------------------------------
|       Admin Panel > Factory Expense Tracker
|-----------------------------------------------------------------------------------------

1. Add New Expense
2. View Detailed Log
3. View Category Summary
0. Back
Enter your Choice...2
Date           Category            Description         Amount
-------------------------------------------------------------------------
21/3           Utility             Rent                Rs. 1500
21/2           Material            Rent                Rs. 3000
21/1           Salary              Rent                Rs. 2000
30/09          Utility             Rent                Rs. 1000
25/12          Utility             Rent                Rs. 50
```

1.4.5 2.2.5 Generate Financial Report (The Core Logic)

1.4.5.1 Purpose of The Menu

To aggregate all data (Work, Attendance, Expenses, Rates) into a professional HTML document that shows the factory's weekly/monthly performance.

1.4.5.2 Input/Output

- **Input:** `Start Day` (Int) and `End Day` (Int).
- **Output:** Generates `Weekly_Financial_Report.html` and opens it in the browser.

1.4.5.3 Validation

Checks that `startDay <= endDay`. It uses the logic we fixed to ensure salaries don't go negative if penalties exceed the earned amount.

```
|-------------------------------------------------------------------------------------------
|       Main menu > Login > Admin Panel > Generate Bill
|-------------------------------------------------------------------------------------------
Enter Start Day for the week (e.g., 1): 1
Enter End Day for the week (e.g., 7): 7

Report updated successfully for Days 1-7!Press Any Key to Continue..
```

-Generate Weekly Financial Report:

Textile Factory Financial Report

Reporting Period: Day 1 to Day 7

Employee Payroll (with Attendance Penalty)

Name	Bundles	Absents	Penalty	Net Salary
Ali	50	0	Rs. 0	Rs. 1500
Awais	67	2	Rs. 400	Rs. 2280
Hamza	56	2	Rs. 400	Rs. 2400

Party Invoices

Party Name	Bundles Dyed	Price/Bundle	Total Amount
BUTT BROTHERS	50	400	Rs. 20000
BILAL THREADS	100	350	Rs. 35000
AMAN TRADERS	23	200	Rs. 4600

Factory Expense Breakdown

Date	Category	Description	Amount
21/3	Utility	Rent	Rs. 1500
21/2	Material	Rent	Rs. 3000
21/1	Salary	Rent	Rs. 2000
30/09	Utility	Rent	Rs. 1000
25/12	Utility	Rent	Rs. 50

Total Revenue: Rs. 59600
Total Salaries (After Penalties): Rs. 6180
Factory Expenses: Rs. 7550

Net Business Profit: Rs. 45870

1.4.6 View Production Summary (Admin View)

1.4.6.1 Purpose of The Menu

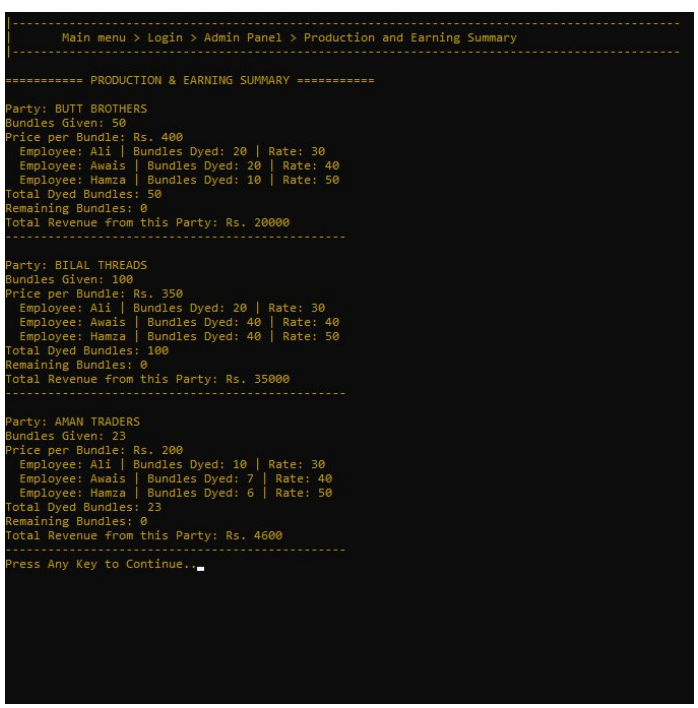
The purpose of this menu is to provide a real-time statistical overview of the factory's output. It allows the user to see the cumulative work performed by a specific employee across all clients (Parties) for the current month. This is the primary verification tool used before final bills are generated.

1.4.6.2 Input/Output

- **Input:** The system automatically detects the logged-in `username` (or the Admin selects an `Employee ID`).
- **Output:** A detailed console table showing:
  - **Party Name:** The specific client for whom the thread was dyed.
  - **Total Bundles:** The sum of all bundles dyed across 31 days.
  - **Status:** An indicator showing if the work has been recorded for the current day.
  - **Grand Total:** The final sum of all bundles dyed by that worker.

1.4.6.3 Validation

- **Empty Record Check:** If no work has been recorded for a specific party, the system displays `0` rather than leaving the field blank or crashing.
- **User Filtering:** For the Worker view, the system applies a filter to ensure they can only access their own production stats, preventing them from seeing the output of other employees.
- **Array Bounds:** The loop is strictly limited to the `partyIndex` to ensure the system doesn't display "garbage" data from empty array slots.



1.5 User Sub-Menu

1.5.1 Enter Daily Work

1.5.1.1 Purpose of The Menu

To allow workers to record the specific number of thread bundles they have dyed for a particular client (Party) on any given day of the month.

1.5.1.2 Input/Output

- **Input:** `Party Index` (Integer selection from the list of clients). * `Day` (Integer, 1-31). * `Bundle Count` (Integer, number of bundles completed).
- **Output:** The data is saved into the `dailyWork` 3D array and a "Work Record Updated" confirmation is displayed on the console.

1.5.1.3 Validation

The system checks that the `Party Index` exists. If a user enters a non-numeric value or an invalid index, the system prompts: "Invalid Party selection. Please try again." It also ensures the `Day` does not exceed 31.

-User Sub-Menu:

1.5.2 View My Weekly Progress

1.5.2.1 Purpose of The Menu

To provide transparency to the worker, allowing them to see a summary of their production for the current week or month and an estimate of their earned wages.

1.5.2.2 Input/Output

- **Input:** The system automatically uses the logged-in `username` to filter data.

- **Output:** A table showing:
  - Total Bundles dyed per Party.
  - Estimated Gross Salary (Bundles × Worker's Rate).
  - Attendance status for the current week.

Validation This is a read-only menu. No input is required, so the main validation is ensuring the correct data is filtered for the specific logged-in worker so they cannot see other employees' data.

-Enter Daily Work:

```
|-------------------------------------------------------------------------------------------
|       Main menu > Login > User Panel  > Enter Daily Work
|-------------------------------------------------------------------------------------------
----Select party---- 0. BUTT BROTHERS
1. BILAL THREADS
2. AMAN TRADERS
Enter party number: 0
Enter Day(1-31): 1
Enter Bundles dyed: 0
Work Saved Successfully.Press Any Key to Continue..
```

1.5.3 Update Profile / Change Password

1.5.3.1 Purpose of The Menu

To allow workers to maintain security by updating their login credentials without needing Admin intervention.

1.5.3.2 Input/Output

- **Input:** `Current Password` (String) and `New Password` (String).

- **Output:** Updates the `users.txt` file and displays "Password Changed Successfully."
-

1.5.3.3 Validation

The system first validates the `Current Password`. If it does not match the stored record, the update is denied with the message: "Incorrect current password. Access denied."

-Update Profile:

```
|-------------------------------------------------------------------------------------------
|       Main menu > Login > User Panel  > Update Profile
|-------------------------------------------------------------------------------------------
Enter New Username: Ahmad
Enter New password: 12341234
Updated successfully! Please re-login with new credentials.Press Any Key to Continue..
```

1.5.4 View Parties

1.5.4.1 Purpose of The Menu

To display a comprehensive and up-to-date list of all clients (Parties) currently registered in the **Thread Dyeing Management System**. This allows the Admin to verify client names and the current dyeing rates (prices) being charged to them.

1.5.4.2 Input/Output

- **Input:** Admin selects the numeric option for "View Parties" from the Party Module menu.
- **Output:** The system outputs a neatly formatted table on the console.

```
-------------------------------------------
ID     | Party Name          | Price/Bundle
-------------------------------------------
0      | Butt Brothers       | Rs. 200
1      | Bilal Textile       | Rs. 250
2      | Aman Fabrics        | Rs. 180
-------------------------------------------
```

1.5.4.3 Validation

The system checks the `partyIndex` variable. If `partyIndex == 0` (meaning no parties have been added yet), the system displays a validation message: *"No parties found. Please add a party first."* This prevents the program from trying to display an empty table.

Here is the documentation for the **Weekly Progress** feature under the User/Worker section. This is a vital part of your report as it demonstrates how the system calculates earnings based on the 3D array data.

1.5.5 2.3.2 View Weekly Progress

Purpose of The Menu The purpose of this menu is to provide workers with a transparent view of their productivity and estimated earnings. It aggregates the data from the `dailyWork` 3D array for a specific 7-day period (or the current month-to-date) and calculates the gross pay based on the worker's assigned piece-rate.

Input/Output

- **Input:** The system automatically identifies the worker via the `username` session variable. The user may input a `Week Number` (1-4) to filter results.
- **Output:** A formatted table showing:
  - **Party Name:** List of clients worked for.
  - **Total Bundles:** Total units dyed for that client during the week.
  - **Piece Rate:** The worker's current pay per bundle (e.g., Rs. 15).
  - **Sub-Total:** Total bundles multiplied by the rate.
  - **Attendance Summary:** Number of days present vs. absent.

Validation

- **Zero-Data Handling:** If a worker has not logged any work for the selected period, the system displays "No records found for this period" instead of empty rows or garbage values.
- **Read-Only Integrity:** This interface is strictly read-only; no inputs are accepted that could modify the database, ensuring that production records remain tamper-proof from this menu.
- **Session Consistency:** Validates that the data being pulled matches the `EmployeeID` linked to the logged-in `UserID`.

-View weekly progress:

```
|-------------------------------------------------------------------------------------------
|       Main menu > Login > User Panel  > View Weekly Progress
|-------------------------------------------------------------------------------------------
Weekly progress for Ali----------------------------BUTT BROTHERS: 20 bundles
BILAL THREADS: 40 bundles
AMAN TRADERS: 10 bundles

Total bundles dyed this week: 70
Press Any Key to Continue..
```

-Weakness in the App: